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A RESEARCH PROGRAMME IN BOUNDED SELF-MONITORING

SĪDUS

Towards Metarepresentational Ecology

DOCUMENT III

The Tractable Horizon

Radboud Master's: One Experiment, Sized to Fit

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specialist · quantity · way

Truth, maximally sought. Ambition tempered by humility.
Beauty tempered by goodness. Honesty above all else.

Sīdus: Towards Metarepresentational Ecology

III. Radboud Master's: The Tractable Horizon

Programme-specific scoping note - August 2026
Status: a Master's-scale research plan, not the full Sīdus programme.

> ***The Master's thesis should test one candidate organizing principle. It should not attempt to establish a metarepresentational ecology.***

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0. Current institutional reality

For the 2026-2027 academic year, Radboud's Artificial Intelligence Master's currently lists **Machine Learning and Neural Computing (MLNC)** and **Human Centered Intelligent Systems (HCIS)** as the offered specialisations. Radboud states that the planned Computational Cognitive Science specialisation will not start in 2026-2027 or the following academic year. [1]

For this programme, **MLNC is the cleaner current home** because the tractable project is computational: artificial agents, partial observability, learned representations, information constraints, causal interventions, and probabilistic/model-comparison methods. Because the user's MSc entry would follow the 2027 pre-master and therefore begin in 2028, this institutional fit should be **re-checked in spring 2027** rather than treated as frozen programme fact.

That does not make the project merely an ML engineering exercise. Radboud's **Artificial Cognitive Systems** group explicitly combines computational modelling rooted in AI and computational neuroscience with empirical neural and behavioural research. [2] The **Computational Cognitive Science** group remains an important intellectual neighbour: it emphasizes conceptual analysis, formal/computational modelling, complexity analysis, and simulations, and includes work on theory building and computational explanation. [3]

The practical fit is therefore:

> ***Primary implementation fit:*** *MLNC + Artificial Cognitive Systems*
> ***Formal/adversarial fit:*** *Computational Cognitive Science*

This is an assessment of research fit, not a claim about supervision availability.

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1. The Master's project

Working title

Internal-State Monitoring Under Scarcity: Causal and Information-Theoretic Tests of Learned Resource Allocation

One-sentence question

> ***When several private internal control states are useful to a coordinating agent but can only be exposed through a genuinely binding shared information-rate channel, what predicts which states the agent learns to communicate, at what rate, and whether those messages are causally used?***

This is the **E1 Scarcity Terrarium** from the research programme.

It intentionally does **not** ask:

- whether the agent is conscious;
- whether all internal-state messages are strict metarepresentations;
- whether human cognition has one global self-monitoring budget;
- whether a metarepresentational ecology exists;
- whether attention is uniquely privileged.

It tests one possible organizing pressure - **scarcity** - in a substrate where ground truth and interventions are available.

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2. Why this is the right tractable horizon

The full programme contains at least four difficult scientific problems:

1. defining genuine metarepresentation rather than generic internal monitoring;
2. discovering whether monitoring systems share resources;
3. identifying causal-topological relations among them;
4. determining whether any of these principles generalize to biological cognition.

A Master's cannot solve all four without becoming a manifesto rather than a thesis.

The chosen project has a better risk profile:

- **internal states are known by construction;**
- **privacy/access can be controlled;**
- **scarcity can be manipulated rather than inferred;**
- **rate and incremental information can be estimated;**
- **messages can be selectively perturbed;**
- **negative results remain interpretable;**
- **no phenomenality measurement problem is imported.**

The thesis is therefore useful even if the larger ecology hypothesis is wrong.

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3. Thesis architecture

Use a compact partially observed control environment with at least three low-cardinality private states:

$$C_{1,t}, C_{2,t}, C_{3,t}.$$

Candidate functions can include:

- sensory selection / fidelity allocation;
- working-memory update/protection;
- action routing or effector selection.

Each state is generated within a local controller. A coordinating policy U receives ordinary information $\mathcal{I}_t$ but cannot directly inspect the private states.

A source-specific encoder may send a compressed internal-state message:

$$C_{v,t} \to S_{v,t} \to U.$$

All sources share a total information-rate constraint:

$$\sum_{v=1}^{K} r_v \le C_{\text{meta}}.$$

The experimenter supplies the candidate source set; the learned agent determines how much rate each source receives. This is **endogenous allocation**, not spontaneous ontology discovery.

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4. Core quantities

4.1 Residual uncertainty

Before receiving the special message, how hidden is the state from ordinary coordinator information?

$$u_v(t) = \frac{H(C_{v,t} \mid \mathcal{I}_t)}{H(C_{v,t})}, \qquad H(C_{v,t}) > 0.$$

If $u_v \approx 0$, the special message has little informational problem to solve.

4.2 Causal control relevance

For small valid state corruption probability δ:

$$c_v(\delta) = \frac{\mathbb{E}[J \mid p_v = 0] - \mathbb{E}[J \mid p_v = \delta]}{\delta}.$$

Use reachable/on-support alternative states rather than arbitrary invalid corruptions.

4.3 Learnability

Estimate how quickly the source can be learned under a dedicated, non-scarce channel. Keep this outside the competitive bottleneck so allocation does not contaminate the covariate used to explain allocation.

4.4 Compressibility

Estimate an empirical distortion-rate or benefit-rate curve for each source. Easy to learn and easy to compress are distinct.

4.5 Rate spent

$$r_v = \mathbb{E}\left[D_{KL}\left(q(S_{v,t} \mid C_{v,t}, \mathcal{I}_t) \| p(S_{v,t} \mid \mathcal{I}_t)\right)\right].$$

Treat this as a variational rate-control surrogate.

[R2 ·correction, revised] The rate–information gap is an identity, not an assumption: $r_v = m_v + \Delta_v^{\text{prior}}$, where the mismatch term is an **expectation over the conditioning variable**, $\Delta_v^{\text{prior}} = \mathbb{E}_{\mathcal{I}}\, D_{KL}(q(S_v \mid \mathcal{I}) \| p(S_v \mid \mathcal{I})) \ge 0$. Report Δ_v^{prior} as a measured quantity alongside r_v and m_v. See Document IV, §3.3a.

4.6 Information delivered

$$m_v = I(C_{v,t}; S_{v,t} \mid \mathcal{I}_t).$$

The central distinction remains:

$$\boxed{r_v \neq m_v.}$$

4.7 Joint benefit surface

$$\mathcal{B}(\mathbf{r}) = \mathbb{E}[J \mid \mathbf{r}] - \mathbb{E}[J \mid \mathbf{0}].$$

The normative experimental benchmark is:

$$\mathbf{r}^*(C) = \arg \max_{\mathbf{r} \geq 0,\, \sum_v r_v \leq C} \mathcal{B}(\mathbf{r}).$$

Do not interpret this as a claim that the learned agent explicitly solves the argmax.

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5. Confirmatory hypotheses

H1 - Binding scarcity produces a real trade-off

As C_{meta} is swept from non-binding to binding capacity, increasing rate to one source should reduce feasible rate available to others and alter performance in predictable ways.

Against H1: nominally lower capacity leaves every source fully informative, or no cross-source displacement occurs despite validated rate control.

H2 - Dynamic control value predicts allocation better than trivial baselines

A joint resource model using measured benefit should predict held-out learned allocations better than:

- uniform allocation;
- source cardinality/entropy alone;
- ordinary inferability alone;
- isolated causal value alone;
- fixed source identity;
- encoder-head identity;
- **myopic value of computation (myopic VOC)** —allocate monitoring rate to source v in proportion to the **one-step** expected improvement in downstream decision quality from resolving C_v (Russell & Wefald 1991; Hay et al. 2012).

[Design addition, August 2026 —revised after release-candidate review] Uniform allocation is a null; entropy-only, inferability-only and isolated-causal-value are **simpler structured models**, not nulls, and an earlier version wrongly lumped all five together. Myopic VOC is added because it is the closest competitor drawn from an established formalism, and because H4's topology hypothesis **overlaps conceptually** with value of computation —graph position is not identical to VOC, but the two are close enough that a topology result could be a VOC result in different notation. The comparator is specifically one-step value and is named as such: **myopic VOC**, one approximation within the metalevel-MDP framework, not VOC in general.

Against H2: the richer resource model does not improve out-of-sample prediction over these baselines. **If it fails specifically against myopic VOC**, the honest reading is that a joint, budget-constrained resource surface adds nothing beyond myopic per-source value of computation. That is a clean, publishable negative result and it should be pre-registered as one.

H3 - Messages are causally used

Selective corruption/replacement of S_v should impair the downstream behavior for which information about C_v is useful, while preserving non-target messages and remaining near trained support.

Against H3: C_v is decodable from S_v but removing/corrupting S_v has no relevant behavioral effect.

H4 - Topology is a candidate secondary predictor

If technically feasible, vary or measure downstream reach/persistence and test whether these properties improve prediction beyond direct causal value and inferability.

This is secondary. It should not endanger completion of H1-H3.

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6. The controls that are not optional

1. **Direct-read oracle.** Give the coordinator direct access to selected private states to estimate the ceiling and verify that hidden-state information can help.
2. **No-seam/full-read control.** Confirm that the communication problem is generated by restricted access rather than the task alone.
3. **Non-binding capacity condition.** All candidate heads must be capable of becoming informative before scarcity is imposed.
4. **Head permutation across seeds.** Source identity must not be confounded with a privileged encoder index.
5. **Scarcity ramp with a real constraint.** Avoid treating a fixed penalty coefficient as equivalent to a known shared budget.
6. **Allocation plateau.** Measure confirmatory rates only after budget, return and rate trajectories stabilize by a preregistered rule.
7. **From-scratch diagnostic for selected conditions.** If ramped and from-scratch training disagree, report optimization hysteresis.
8. **Held-out environments/contingencies.** The allocation model should predict more than the exact configurations used to fit it.

These controls are the scientific core. They are more important than decorative model complexity.

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7. What is deliberately excluded from the core thesis

No primary attention residual

The earlier Sīdus question asked whether attention received excess rate relative to a resource baseline. In the new programme, attention is one source among others. Generic residuals may be inspected, but the thesis does not depend on a special attention effect.

No strict-metarepresentation claim

The messages in this experiment are **internal-state models**. That terminology avoids pretending the project has already solved the contested definition of metarepresentation.

No causal abstraction by default

Interchange-intervention/causal-abstraction analysis is elegant but high overhead. Use it only if a robust role-dependent effect has already been established and the high-level role claim materially matters.

No Gaussian-process Bayesian optimization by default

Start with a small factorial/simplex design. Escalate to Astrocyte only if training evaluations become sufficiently expensive that active surface estimation is justified.

No human or clinical translation

A human study should not be added as a thesis appendix merely to make the work look neuroscientific. Biological translation becomes meaningful after construct validity and a reproducible computational phenomenon exist.

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8. One optional extension - choose at most one

If H1-H3 are completed early and cleanly, choose **one** extension.

Option A - strict metarepresentation benchmark

Add a source whose target is a representational property/process, such as a hidden reliability/source regime affecting a first-order perceptual representation. Test whether a meta-process model beats matched first-order/public-context alternatives.

Why this option is attractive: it ties the thesis more directly to the broader programme while preserving a clear computational manipulation.

Option B - recursive allocator monitor

Add a first-order resource allocator with C_{base} and a separate monitoring budget C_{meta}. Compare feedback-enabled and frozen-feedback conditions.

Why this option is attractive: it retains the strongest S1-R idea and connects to meta-control.

Option C - causal topology

Manipulate downstream reach/persistence for one source and ask whether benefit/allocation follows causal position.

Why this option is attractive: it tests a candidate organizing principle directly.

Rule: completing the core thesis beats attempting two extensions.

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9. Stage gates and schedule for a 45 EC extended research project

Radboud's current MLNC curriculum allows either a 30 EC research project plus 15 EC internship or a **45 EC extended research project**. [4] The latter is the cleaner fit for a deep computational thesis if available and approved.

The timeline below is an illustrative research schedule, not an official Radboud timetable.

Stage	Illustrative project window	Deliverable	Stop/narrow decision
Prior art + construct freeze	month 1	concise literature/claim register; frozen core question	if E1 is already duplicated exactly, redesign before coding
Environment + baseline	months 1-2	compact POMDP; working local controllers; direct-read baseline	if private states do not matter to performance, redesign task
Information estimators	month 2	validated r_v, m_v, u_v measurements in toy ground-truth cases	if estimates are unstable, simplify targets
Non-binding communication	months 2-3	all heads demonstrably informative	if heads cannot learn, do not impose scarcity
Scarcity pilot	months 2-3	stable rate-control manipulation and preliminary trade-offs	**Gate:** if no real trade-off, narrow/stop scarcity claim
Freeze + preregister analysis	end of month 3	fixed environment, manipulations, models and seed/precision plan	no confirmatory tuning afterward
Confirmatory E1	months 4-6	held-out allocation/return data across seeds	if results are path-dependent, report that rather than hiding it
Causal-use + topology secondary	months 6-7	selective interventions; optional topology test	drop secondary analyses if core is not complete
Optional single extension	month 7	E2 *or* recursive *or* topology extension	optional, never required for success
Replication + writing	months 8-9	thesis, code, data, negative-result ledger	finish before expanding

10. Radboud curriculum alignment

Radboud lists MLNC as a 120 EC Master's. The 2026-2027 provisional structure includes 18 EC common compulsory courses, 18 EC specialisation courses, 21 EC track courses, 18 EC free electives, and either the 45 EC extended research project or a 30 EC project plus 15 EC internship. [4]

The listed MLNC specialisation courses are particularly aligned with the methods this thesis needs:

- **Bayesian Machine Learning** - uncertainty, model comparison, posterior reasoning;
- **Complex Adaptive Systems** - interacting modules and system-level dynamics;
- **Neural Computation** - computational-neuroscience grounding;
- **Advanced Deep Learning** - implementation and representation learning. [4]

For the **Machine Learning track**, Radboud lists Probabilistic Deep Learning among the compulsory track courses; that track is the cleaner methodological match if the thesis remains focused on probabilistic representation, bottlenecks and learned agents. [4]

The **Neural Computing track** is more appropriate if the project shifts toward neuromorphic systems, BCI or biologically constrained implementation. Radboud describes that track as focused on brain-inspired technologies, including neuromorphic systems and brain-computer interfaces. [4]

11. Pre-Master's as preparation, not administrative delay

Radboud's 2026-2027 page for a Life/Social Sciences background lists a 57 EC MLNC pre-Master's. It includes Calculus 2, Probability Theory, Programming 1 and 2, From Data to Model, Linear Algebra, Bayesian Networks, Reinforcement Learning, Deep Learning, introductory digital signal processing, Brain-Computer Interfacing, Bayesian Statistics, and Cognitive Computational Neuroscience. [5]

That sequence maps unusually well onto the thesis's actual dependencies:

$$\begin{aligned}\text{probability} &\longrightarrow \text{information theory} \longrightarrow \text{optimization}\\ &\longrightarrow \text{RL/POMDPs} \longrightarrow \text{probabilistic modelling}\\ &\longrightarrow \text{causal intervention.}\end{aligned}$$

The gap is information theory/rate-distortion itself, which should be studied independently alongside the programme.

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12. Research-group fit

Artificial Cognitive Systems - primary implementation fit

Radboud describes the Artificial Cognitive Systems group as combining computational modelling rooted in AI and computational neuroscience with empirical neural and behavioural research; the group is led by Marcel van Gerven. [2]

Why the thesis fits:

- learned artificial cognitive systems;
- neural/computational modelling;
- probabilistic representation and control;
- a plausible future bridge to empirical neural/behavioural data;
- complete internal access in the synthetic phase.

Computational Cognitive Science - formal/adversarial fit

The Computational Cognitive Science group, led by Iris van Rooij, explicitly uses conceptual analysis, formal/computational modelling, complexity analysis and simulation across symbolic, neural, Bayesian, dynamical and logical approaches. [3] The group lists Guest & Martin's work on computational modelling as theory-forcing and van Rooij & Baggio's work on theory before testing among its key publications. [3,6,7]

Why that matters:

- the programme is especially vulnerable to formal-looking but unconstraining models;
- "metarepresentational ecology"needs a defined explanandum, not merely an appealing metaphor;
- normative allocation models need a computational-level interpretation distinct from learned implementation;
- tractability and identifiability should constrain claims before large experiments are run.

The CCS **research group exists regardless of the delayed Master's specialisation**. The appropriate claim is intellectual proximity, not an assumption that a particular researcher will supervise the project.

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13. What to bring to a prospective supervisor

Bring a **two-page study**, not the whole Sīdus history.

The supervisor-facing package should contain:

1. one paragraph of prior art;
2. one computational question;
3. one architecture diagram;
4. H1-H3;
5. the direct-read/no-seam/scarcity controls;
6. the estimands r_v, m_v, $\mathcal{B}(\mathbf{r})$;
7. the primary comparison models;
8. one paragraph of kill criteria;
9. a compute pilot plan.

Questions worth asking:

- Is the shared-rate competition experiment already present in a literature we have missed?
- Is the proposed information seam a useful abstraction for this lab, or does it bake in too much modularity?
- What simpler baseline would you demand before accepting the joint resource model?
- Would you consider the artificial-agent result cognitively explanatory, or only an engineering demonstration?
- Which part should be removed to fit a Master's timeline?
- Which lab infrastructure or existing environment could make the test cheaper and more biologically meaningful?

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14. What not to say in a Master's proposal

Do not say:

- "I am developing a theory of metarepresentational ecology."
- "The brain has a global metarepresentational budget."
- "This model explains consciousness."
- "Attention is uniquely worth monitoring."
- "The model discovers its own self."
- "The glyphs represent deep computational principles."

Say:

> ***I want to test whether a binding information-rate constraint produces predictable competition among models of private internal control states in a modular learning agent, and whether the resulting messages are causally used. This is one candidate mechanism that could later inform a broader construct-first account of internal monitoring.***

That is enough.

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15. Success conditions

The Master's is successful if any one of the following is established cleanly:

- a real shared rate constraint produces stable, reproducible allocation trade-offs;
- a joint benefit model predicts learned allocation out of sample;
- a simpler model explains everything and the larger allocation story is unnecessary;

- rate allocation is strongly path-dependent, showing that the apparent "law"is an optimization history;
- transmitted internal-state information is decodable but behaviorally unused;
- the information seam proves unnecessary in the tested task;
- a controlled topology manipulation changes monitoring value;
- the experiment falsifies a central assumption and thereby prevents the larger programme from building on it.

The worst outcome is not a null result.

The worst outcome is an ambiguous system with too many moving parts to know what failed.

16. The tractable horizon

The Master's should end with one of two conclusions.

Positive

> ***Under controlled partial access and a binding information-rate constraint, learned internal-state monitoring exhibits a reproducible allocation principle that survives simpler baselines and causal-use tests. This justifies testing whether the same principle applies to stricter metarepresentations or other architectures.***

Negative

> ***The tested scarcity formulation does not produce a stable/generalizable allocation principle, or its effects reduce to simpler inferability, implementation or optimization factors. The broader programme should not treat scarcity as an organizing law without a different empirical basis.***

Both conclusions move the programme forward.

A one-paper thesis that kills a grander idea remains a successful scientific outcome.

References and current Radboud sources

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